

Title

Project Planning – SB Stocks

Project Objective

The objective is to design and develop a virtual stock trading platform that provides practical stock market simulation.

Project Phases

Phase 1 – Brainstorming and Ideation

Activities:

- Identify user problems.
- Generate solution ideas.
- Prioritize ideas.
- Define the problem statement.

Phase 2 – Requirement Analysis

Activities:

- Identify functional requirements.
- Identify non-functional requirements.
- Analyze user journey.
- Design data flow.
- Select technologies.

Phase 3 – Project Design

Activities:

- Design solution architecture.
- Define database entities.
- Design application flow.
- Plan frontend and backend modules.

Phase 4 – Development

Activities:

- Configure React frontend.

- Configure Express backend.
- Configure MongoDB.
- Develop authentication.
- Develop stock APIs.
- Implement portfolios.
- Implement trading.
- Implement history.
- Implement admin module.
- Implement charts.

Phase 5 – Testing

Activities:

- API testing.
- Authentication testing.
- Trading validation.
- Performance testing.
- Frontend testing.

Phase 6 – Documentation

Activities:

- Prepare project documentation.
- Document architecture.
- Document execution steps.
- Prepare project files.

Development Milestones

Milestone	Deliverable
M1	Problem Definition
M2	Requirement Analysis
M3	Architecture Design
M4	Database Configuration
M5	Authentication
M6	Stock Module
M7	Portfolio Module
M8	Trading Module
M9	Admin Module
M10	Testing
M11	Documentation

Risk Management

Risk	Mitigation
Market API unavailable	Demo data fallback
Invalid user input	Backend validation
Unauthorized access	JWT authentication
Overselling stocks	Holdings validation
Insufficient balance	Balance validation
Database errors	Error handling
UI API failure	Error and loading states

Conclusion

The project planning process divides SB Stocks into manageable development phases and ensures structured implementation.